Practical 14 – Packet Sniffing Using RAW Sockets

AIM

To use raw sockets to capture network packets and display basic packet details.

ALGORITHM

1. Import Python's socket module.
2. Open a raw socket with the required address family and protocol.
3. Run the program with the necessary system permissions.
4. Receive packets available on the selected network interface.
5. Display each packet's source address and length.
6. Stop the capture and close the socket.

PYTHON CODE – WINDOWS

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
s.bind(("0.0.0.0", 0))
s.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
s.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet capture started...")
try:
    while True:
        data, addr = s.recvfrom(65535)
        print("Packet received from:", addr[0])
        print("Packet length:", len(data))
        print("-" * 40)
except KeyboardInterrupt:
    print("\nCapture stopped.")
finally:
    s.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    s.close()
```

INPUT

Packets available to the selected network interface on the local computer.

OUTPUT

For each captured packet, the program displays the source IP address and the packet size in bytes.

NOTE

This example uses Windows-specific raw-socket functionality and normally requires Administrator privileges. Capture only traffic you are authorized to inspect.

RESULT

A basic raw-socket packet capture program was prepared to display packet source addresses and lengths.